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AI for Bharat Hackathon

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Team Name :RUGVED SYSTEMS

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Problem Statement :Build an AI-powered solution that improves access to information, resources, or opportunities for communities and public systems.

The Challenge

Critical barriers prevent citizens from accessing government benefits

Low Awareness

30-40%

of eligible beneficiaries are aware of schemes they qualify for

High Drop-off

70%

application abandonment rate due to complexity and confusion

Language Barriers

22

official languages with limited digital support for regional languages

Fragmentation

100+

government portals with scattered information and no unified access

⚠ **₹50,000+ Crores** in scheme budgets remain unclaimed annually

Our Solution: Bharat Access Hub

AI-powered platform bridging the information access gap

Conversational AI Interface

Natural language chatbot powered by RAG (Retrieval Augmented Generation) that understands context, answers questions, and guides users through complex scheme information in their preferred language.

Intelligent Eligibility Matching

AI-driven scoring algorithm that analyzes user profiles against scheme criteria, providing personalized eligibility scores (0-100) and ranking opportunities by relevance and benefit value.

Document Intelligence

OCR-powered form processing with Amazon Textract that extracts text, explains fields in simple language, auto-fills from user profiles, and validates submissions before sending to government portals.

Platform Impact

10,000+
Concurrent Users

<3 sec
Response Time

95%
Matching Accuracy

10
Indian Languages

Security First

End-to-end encryption with AWS KMS, DPDPA 2023 compliant, RBAC access control, and comprehensive audit logging.

What Makes Us Different

Unique capabilities that set Bharat Access Hub apart

RAG-Powered Multilingual Chatbot

Unlike simple FAQ bots, our RAG system retrieves real-time scheme data and generates contextual responses in 10 Indian languages using Claude 3 Sonnet.

- Context-aware conversations

AI-Driven Eligibility Scoring

Proprietary algorithm computes 0-100 eligibility scores by matching user profiles against scheme criteria with weighted factors for demographics, income, location, and more.

- Personalized recommendations

Intelligent Document Processing

Amazon Textract OCR extracts form fields, explains them in simple language, auto-fills from user profiles with 90%+ accuracy, and validates before submission.

- Reduces form completion time by 70%

Voice Interface for All

Amazon Transcribe and Polly enable voice-based interactions for low-literacy users, supporting 10 languages with adjustable speech rates for better comprehension.

- Accessibility-first design

Unified Portal Integration

Seamless integration with 100+ government portals, providing direct application links with pre-filled data and tracking application status across multiple systems.

- One platform, all schemes

Enterprise-Grade Security

End-to-end encryption with AWS KMS, DPDPA 2023 compliance, RBAC access control, comprehensive audit logging, and user data export/deletion rights.

- Privacy by design

Key Features

Six core capabilities powering the platform

Smart Profile Builder

Progressive questionnaire capturing demographics, location, education, employment, income, and family details with real-time completion tracking.

Progressive | Auto-save

AI Chatbot with RAG

Conversational AI powered by Amazon Bedrock Claude 3 Sonnet with RAG pipeline for accurate, contextual responses about schemes and processes.

10 Languages | RAG

Eligibility Engine

AI-driven scoring algorithm matching user profiles against scheme criteria with 0-100 scores, ranking by eligibility × benefit × urgency.

0-100 Score | Real-time

Document Intelligence

Amazon Textract OCR extracts text from forms, explains fields in simple language, auto-fills from profiles, and validates submissions.

OCR | Auto-fill

Personalized Dashboard

Curated scheme recommendations ranked by relevance, AI-generated insights, urgent deadline alerts, and application status tracking.

AI Insights | Tracking

Voice Interface

Amazon Transcribe and Polly enable voice-based interactions for low-literacy users with adjustable speech rates in 10 languages.

Speech-to-Text | Text-to-Speech

User Journey Flow

End-to-end experience from registration to application

Registration
Cognito Auth

Profile
DynamoDB

Discovery
AI Chatbot

Eligibility
Scoring

Documents
OCR + Fill

Apply
Portal Link

Track
Status

Time to First Scheme

Users see personalized recommendations within **5 seconds** of profile completion

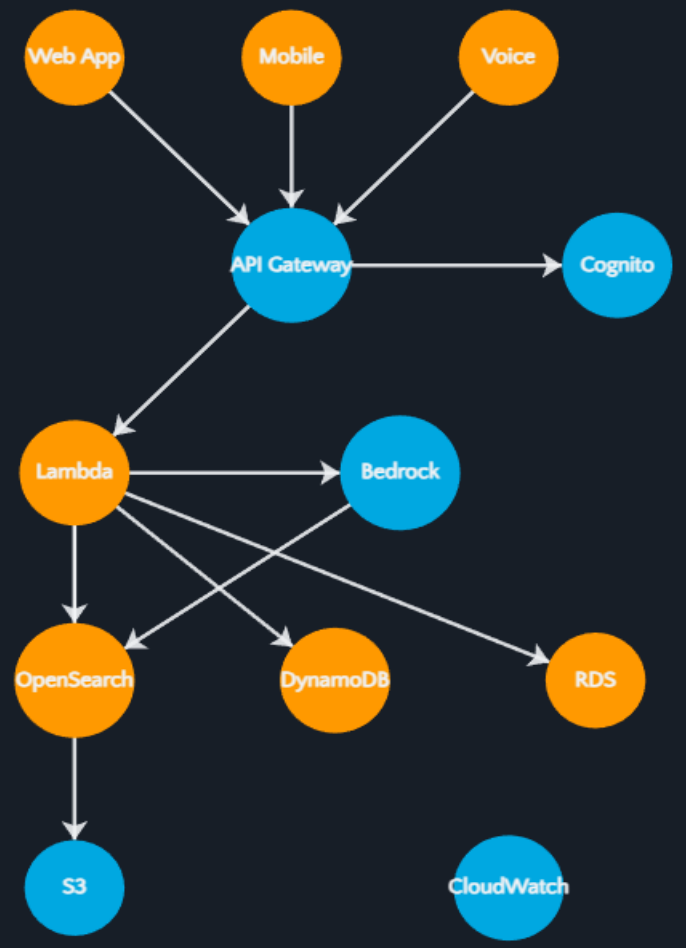
Application Success Rate

Document assistance and validation improve completion rates from **30% to 85%**

Continuous Updates

Eligibility scores auto-recalculate when profile changes or new schemes launch

System Architecture



Client Layer

React Web App, React Native Mobile, Voice Interface via CloudFront CDN

API & Security

API Gateway (REST + WebSocket), AWS Cognito Auth, RBAC, WAF protection

Lambda Services

Profile, Eligibility, Chat, Document, Search, Notification microservices

AI/ML Services

Bedrock (Claude 3, Titan), Textract, Comprehend, Transcribe, Polly, Translate

Data Layer

OpenSearch (Vector DB), DynamoDB, RDS PostgreSQL, S3, ElastiCache Redis

RAG Pipeline - The AI Brain

RAG Data Flow

Document
PDF/URL

→

Chunking
512 tokens

→

Embeddings
1536-dim

→

OpenSearch
Vector DB

User Query
Any Language

→

Semantic
Top 5 Chunks

→

Claude 3
Generate

→

Response
Personalized

Document Chunking

Chunk Size:	512 tokens
Overlap:	50 tokens
Boundary:	Sentence

Titan Embeddings

Dimensions:	1536
Max Input:	8192 tokens
Cost:	$0.10/M tokens

Claude 3 Sonnet

Context:	200K tokens
Input Cost:	$3/M tokens
Output Cost:	$15/M tokens

<3s	95%	10	40%+
Response Time	Accuracy	Languages	Cache Hit

AWS Technology Stack

AI / ML

Amazon Bedrock
Claude 3 Sonnet LLM, Titan Embeddings

Amazon Textract
OCR, form field extraction

Amazon Comprehend
NLP, language detection

Transcribe & Polly
Speech-to-text, text-to-speech

Amazon Translate
Real-time translation, 10 languages

Compute & API

AWS Lambda
Serverless microservices, Python/Node.js

API Gateway
REST + WebSocket APIs

Amazon Cognito
User auth, MFA, JWT tokens

AWS WAF
DDoS protection, rate limiting

CloudFront
Global CDN, edge caching

Data & Storage

Amazon OpenSearch
Vector DB, semantic search, 1536-dim

Amazon DynamoDB
NoSQL, users, chat history, applications

Amazon RDS
PostgreSQL, scheme metadata

Amazon S3
Document storage, static assets

ElastiCache
Redis caching, session store

CloudWatch - Monitoring & Logging | **AWS KMS** - Encryption | **X-Ray** - Distributed Tracing

Impact & Metrics

Projected performance and societal impact

Concurrent Users

10,000+

Serverless auto-scaling handles traffic spikes seamlessly

Response Time

<3 sec

95% of chatbot queries answered within 3 seconds

Societal Impact

₹50,000+ Crores
Unclaimed benefits accessed annually

Millions of Citizens
Empowered with scheme awareness

3x Improvement
In application completion rates

Matching Accuracy

95%

Eligibility scoring accuracy with AI-driven criteria evaluation

Translation Savings

70%

Cost reduction through DynamoDB translation caching

System Reliability

Uptime SLA	99.9%
Cache Hit Rate	>40%
Auto-scaling	Instant

Implementation Roadmap

1 Phase 1: MVP

Months 1-3

- **Core Chatbot**
  RAG pipeline with Claude 3, 5 languages
- **Basic Profile Builder**
  Progressive questionnaire, DynamoDB storage
- **Eligibility Engine**
  Scoring algorithm, 100 top schemes
- **Dashboard**
  Personalized recommendations, basic UI

Target Users: 1,000 beta

2 Phase 2: Scale

Months 4-6

- **Document Intelligence**
  Textract OCR, auto-fill, validation
- **Voice Interface**
  Transcribe + Polly, 10 languages
- **Advanced Search**
  Semantic + keyword hybrid, filters
- **Mobile App**
  React Native, offline mode

Target Users: 10,000

3 Phase 3: Optimize

Months 7-12

- **Portal Integrations**
  50+ government portal APIs
- **Advanced Analytics**
  User behavior, scheme performance
- **AI Improvements**
  Fine-tuned models, feedback loop
- **Multi-State Launch**
  Pan-India rollout, regional partners

Target Users: 100,000+

Why AWS?

Comprehensive AI/ML Stack

Amazon Bedrock provides unified access to Claude 3 Sonnet and Titan Embeddings, eliminating the need for multiple AI providers. Textract, Comprehend, Transcribe, Polly, and Translate complete the AI ecosystem.

Serverless Scalability

AWS Lambda auto-scales from 0 to 10,000+ concurrent users instantly. API Gateway handles millions of requests with built-in throttling and caching. Pay only for what you use.

Enterprise Security

AWS KMS provides encryption at rest, Cognito handles authentication with MFA support, WAF protects against attacks, and comprehensive audit logging ensures DPDPA 2023 compliance.

Multi-Region Availability

Deploy across Mumbai, Hyderabad, and other regions for low-latency access. CloudFront CDN delivers content from 450+ edge locations. 99.9% uptime SLA with multi-AZ redundancy.

Cost-Effective Model

Pay-per-use pricing aligns costs with actual usage. Reserved capacity for predictable workloads. Free tier for development and testing. Estimated monthly cost: ₹2-3 lakhs for 10K users.

200+ Services Ecosystem

Seamless integration between services. OpenSearch for vector search, DynamoDB for NoSQL, RDS for relational data, S3 for storage, and 195+ more services for future expansion.

Empowering Bharat Through AI

Bharat Access Hub bridges the gap between citizens and government services,
one conversation at a time.

10	100+	₹50K Cr+
Indian Languages	Portal Integrations	Benefits Unlocked

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Built for Bharat

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Thank You